

An Archaeological Inference Problem Continued

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Review

- **Site:** Danebury Iron Age hill fort, England
- **Goal:** Estimate the start and end dates of ceramic phases of artifacts from the site by creating a statistical model
- **Challenges:** There are difficulties with mapping radiocarbon dates to actual calendar dates and with getting accurate radiocarbon dates

Review of the Calibration Curve

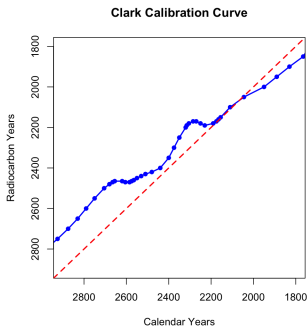
We have a calibration curve $\mu(\theta)$ that maps radiocarbon dates to calendar dates, modeled as a piecewise linear function.

The calibration curve $\mu(\theta)$ is defined by a set of $K + 1$ knot points $((t_k, m_k), k = 0, 1, \dots, K)$. Here, t_k represents calendar years (θ), and m_k corresponds to radiocarbon years (μ). Between two adjacent knots, the relationship is piecewise linear:

$$\begin{aligned}\mu(\theta) &= a_k + b_k(\theta), \quad t_{k-1} < \theta \leq t_k, \\ a_k &= m_k - b_k t_k, \quad b_k = \frac{m_k - m_{k-1}}{t_k - t_{k-1}} \neq 0.\end{aligned}$$

Calibration Curve

- Used to estimate the calendar dates corresponding to the radiocarbon dates of ceramic artifacts



Constructing the Likelihood Function

We want to compute the probability of observing radiocarbon dates x given the ceramic phase j .

$$p(x|s, j, \alpha) = \int_{\alpha_{j+1}}^{\alpha_j} p(x|\mu(\theta), s)p(\theta|\alpha_j, \alpha_{j+1})d\theta \quad (1)$$

The likelihood for each observation (x, s, j) integrates over all possible calendar dates θ within the ceramic phase $(\alpha_j, \alpha_{j+1}]$.

Building the Log-Likelihood Function

Our goal is to estimate the phase boundaries,

$$\alpha = (\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5),$$

based on the radiocarbon measurements. Each data point x_i is associated with:

- a measurement error s_i
- a phase index j_i , meaning it falls between α_{j_i} and α_{j_i+1}

The structure relies on choosing two indices $l_1(j), l_2(j)$ so that:

$$t_{l_1(j)-1} < \alpha_{j+1} \leq t_{l_1(j)} < \cdots < t_{l_2(j)} \leq \alpha_j < t_{l_2(j)+1}.$$

Piecewise Linear Approximation

To account for calibration uncertainty, we integrate over all possible calendar dates θ that a radiocarbon measurement between α_{j_i} and α_{j_i+1} , using the calibration function $\mu(\theta)$. To avoid directly integrating over the complex calibration curve, we approximate the integral as a sum over small linear segments:

$$p(x|s, j, \alpha) = \frac{1}{\alpha_j - \alpha_{j-1}} \sum_{i=l_1(j)}^{l_2(j)+1} I_i$$

where,

$$I_i = \int_{d_{i-1}}^{d_i} p(x|a_i + b_i\theta, s) d\theta$$

Each I_i corresponds to a small interval where the calibration curve is linear

The Log-Likelihood Function

The total likelihood of observing the data given the phase boundaries α is the product of individual conditional densities. For computation and optimization, we take the logarithm to obtain the log-likelihood:

$$L(\mathbf{x}; \alpha) = \sum_{i=1}^n \log p(x_i \mid s_i, j_i, \alpha)$$

This converts the product into a sum, and makes the expression easier to maximize.

Implementation in R: Knots

The calibration curve is divided into segments between the knots t_K . We define the piecewise linear function $\mu(\theta)$ to implement the calibration curve:

$$\begin{aligned}\mu(\theta) &= a_k + b_k\theta, & t_{k-1} < \theta \leq t_k, & \quad k = 1, \dots, K, \\ &= a_1 + b_1\theta, & \theta \leq t_0, \\ &= a_K + b_K\theta, & \theta > t_K,\end{aligned}$$

We set boundary conditions to handle the cases where θ is outside the range of given knots.

- returns the interpolated value $\mu(\theta)$

Likelihood Calculation for a Single Observation

Now, we compute the likelihood of a radiocarbon date x coming from a true calendar date θ between phase bounds α_j and α_{j+1} . We integrate over all θ within the phase:

$$\mathcal{L}(x \mid s, \alpha_j, \alpha_{j+1}) = \int_{\alpha_j}^{\alpha_{j+1}} \underbrace{f(x \mid \mu(\theta), s)}_{\text{normal}} \cdot \underbrace{f(\theta \mid \alpha_j, \alpha_{j+1})}_{\text{uniform}} d\theta$$

```
likelihood_x <- function(x, s, alpha_j, alpha_j1) {  
  val=x  
  std=s  
  lb=alpha_j  
  ub=alpha_j1  
  
  p_theta <- function(theta, val.=val, std.=std, lb.=lb, ub.=ub) {  
    mu <- mu_theta(theta) # calibrated mean radiocarbon date for calendar year  
    p_x_given_mu_s = dnorm(val., mean=mu, sd=std.) # normal likelihood  
    p_theta_given_alpha <- dunif(x=theta, min=lb., max=ub.) # uniform prior  
    return(p_x_given_mu_s * p_theta_given_alpha)  
  }  
  integrate(p_theta, lower = lb, upper = ub, val=val, std=std)$value  
}
```

Likelihood for All Observations in a Phase

$$L(\mathbf{x}; \boldsymbol{\alpha}) = \sum_{i=1}^n \log p(x_i \mid s_i, j_i, \boldsymbol{\alpha})$$

For each phase, calculate the likelihood of its dates falling within that phase's computed interval with the constraint that the phases cannot overlap.

```
negll=function(alpha_1,len_4,len_3,len_2,len_1)
{
  l4=mapply(likelihood_x, phase4$x, phase4$s, MoreArgs = list(alpha_j = alpha_1, alpha_j1 = alpha_1+len_4))
  l3=mapply(likelihood_x, phase3$x, phase3$s, MoreArgs = list(alpha_j = alpha_1+len_4, alpha_j1 = alpha_1+len_4+len_3))
  l2=mapply(likelihood_x, phase2$x, phase2$s, MoreArgs = list(alpha_j = alpha_1+len_4+len_3, alpha_j1 = alpha_1+len_4+len_3+len_2))
  l1=mapply(likelihood_x, phase1$x, phase1$s, MoreArgs = list(alpha_j = alpha_1+len_4+len_3+len_2, alpha_j1 = alpha_1+len_4+len_3+len_2+len_1))
  -(sum(log(l4))+sum(log(l3))+sum(log(l2))+sum(log(l1)))
}
```

α_1 is the start of phase 4, while other parameters represent the length of each phase

Maximum Likelihood Estimation

We use the MLE to find the parameter values (phase boundaries) that make the observed data most probable:

```
mle_joint = mle(negll,  
  start = list(alpha_1 = 1850, len_4 = 200, len_3 = 200, len_2 = 200, len_1 = 200),  
  lower = c(1800, 0, 0, 0, 0),  
  upper = c(1900, 300, 300, 300, 300))
```

- Tries many values of α_1 , len_4 , len_3 , etc., starting from our initial guesses
- Computes the likelihood for each combination by summing the log-likelihoods across all phases
- finds the combination that minimizes the negative log-likelihood for each phase

This gives us the most likely calendar intervals for each phase, based on the observed data and the calibration curve.

Challenges with Maximum Likelihood

The authors encountered challenges in relying solely on a Maximum Likelihood Estimation (MLE) to find α .

- numerical optimization often led to different local maxima of the likelihood function, making it difficult to find the global maximum and retain reliable standard error estimates.

The authors instead opted for a Bayesian approach to infer the parameters and the archaeological phases.

Inferring Phases from the Posterior Distribution

Posterior Density: probability density function representing the conditional probability distribution of the unknown parameters given the observed data. It is obtained by combining the likelihood function with the prior distribution using via Bayes' theorem.

Bayes' Theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

We form the posterior density:

$$p(\boldsymbol{\alpha} \mid x) = \frac{e^{L(\mathbf{x}; \boldsymbol{\alpha})} p(\boldsymbol{\alpha})}{p(x)}$$

Posterior Inference in Practice

From the posterior density $p(\alpha \mid \mathbf{x})$, they estimate:

- Posterior means for each α_j
- Posterior standard deviations and intervals
- Posterior covariance between parameters

This provides a full picture of uncertainty and dependence across parameters, which the MLE could not reliably give.

Results

Table: Posterior Means, Standard Deviations for Phase Boundaries (α_i)

	α_1	α_2	α_3	α_4	α_5
Means	2500	2390	2340	2270	1920
Standard Deviations	70	30	40	40	70

Table: MLE Phase Boundaries

	α_1	α_2	α_3	α_4	α_5
Means	2609	2388	2251	2200	1900

Comparing Results

